

First Mid-Semester Exam PHY301 (20 marks)

1. Consider the circuit equation for current I in a LCR circuit:

$$L\ddot{I} + R\dot{I} + I/C = 0$$

Given $L, C > 0$, what is the nature of the fixed point(s) as R is varied ($R \geq 0$).

(5 marks)

2. Consider a conservative Newtonian system where a particle of mass $m = 1$ moves in a potential

$$V(x) = \frac{1}{2}x^2 - \frac{1}{4}x^4$$

- (a) Enumerate the equilibrium (fixed) point(s) of the system.
- (b) Find the nature of the fixed point through analysis of the Jacobian.
- (c) Draw the phase portrait of the dynamics, and mark the separatrix.
- (d) What are the distinct types of motion that can take place in the system?

(8 marks)

3. Consider a system given by the equation: $\ddot{x} + 2\lambda\dot{x} + \omega^2x = F$, where F is a constant.

Find the solution $x(t)$ for the system for the case of $\lambda = \omega$, with initial condition $x(0) = 0$, $\dot{x}(0) = 1$.

(7 marks)